

Supplementary Information

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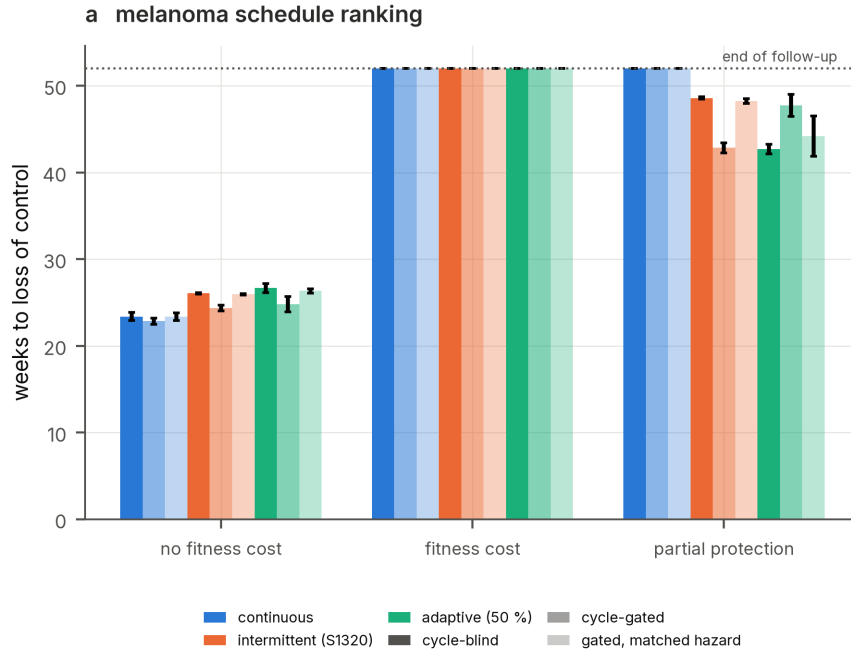
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Whether treatment holidays help depends on three measurable properties of resistant cells: a single-cell simulation of two clinical dosing trials. Supplementary Notes 1–4, Supplementary Figure 1 and Supplementary Tables 1–2.

Supplementary Note 1: Cell-cycle-dependent killing in the melanoma schedules

The death hazard used here has no cell-cycle dependence. The companion Article shows that the cisplatin fate data cannot decide whether cells are spared because they inherited a protective state or because they were outside the replication window, so the same question applies to the schedule comparisons: do their rankings rest on that choice? We repeated the melanoma study with three quarters of the death hazard confined to a window around mid-cycle ($\beta = 0.25$, $\varphi_0 = 0.5$, $w = 0.15$), leaving everything else unchanged, and again with the maximal hazard raised so that the cycle-averaged hazard matches the cycle-blind one.

The window position is a modelling choice rather than a claim about the agent. The melanoma study is built on a BRAF/MEK inhibitor, which does not create replication-coupled lesions as a platinum drug does, although it does act on cells traversing the cycle. What these runs test is robustness to a strong cell-cycle gate of any kind, not the phase specificity of a particular drug.



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20 **Supplementary Fig. 1 | The melanoma schedules under cell-cycle-gated killing.** a, Weeks after
 21 randomisation to loss of control for the three mechanisms and the three clinical schedules (mean $\pm$ s.d. of four
 22 seeds, censored at the end of follow-up), cycle-blind (solid), cycle-gated (mid) and gated at matched hazard (pale).
 23 In the melanoma study the best schedule is unchanged in 3 of 3 mechanisms and the model still favours continuous
 24 dosing under partial protection (52 against 43 weeks, from 52 and 49); the full ordering of the three schedules
 25 survives in 2 of 3 mechanisms (under partial protection the two losing schedules trade places). In the melanoma
 26 study the best schedule is unchanged in 3 of 3 mechanisms at the matched hazard as well.

27 **Supplementary Note 2: Sensitivity of the melanoma schedules to the** 28 **memory of the resistant state**

29 The melanoma study gives the resistant state a memory of five net population doublings, about 20 weeks, which
 30 is a modelling choice and not a measurement, and it is the quantity that decides whether resistant cells revert
 31 during a three-week holiday. Repeating the schedule comparison over memories from 2 to 12 net doublings, with
 32 3 seeds at each point, leaves the ranking in place in the two arms it covers, without and with a fitness cost
 33 of resistance: without a fitness cost both interrupted schedules keep control longer than continuous dosing at
 34 every memory tested. What changes is how much the holidays are worth. The advantage of the intermittent
 35 schedule over continuous dosing falls from 9 weeks at a memory of 2 doublings to 1.7 weeks at 12, against 2.3
 36 weeks at the 5 doublings used in the main text, because a state that is forgotten quickly is re-drawn during
 37 a holiday while one that is remembered is carried through it. The memory assumed here therefore sits at the
 38 conservative end: a shorter memory would make the case for holidays stronger, not weaker. With a fitness cost of
 39 resistance no schedule loses control within the follow-up at any memory in this range, so that arm is insensitive
 40 to the assumption. The scan covers those two arms only. The third arm of the schedule comparison, in which
 41 resistant cells are only partly protected and grow at half speed under drug, is the arm that reproduces the ranking
 42 of SWOG S1320, with continuous dosing keeping control longest; it was not scanned, so how far that ranking

43 depends on the assumed memory is not established here. The main text quotes this arm as 23 weeks against 26,
44 so differencing the rounded times gives 3 weeks; the unrounded difference over the 4 seeds of that run is 2.7 weeks
45 with a standard error of 0.2 weeks, against 2.3 weeks over the 3 seeds of this scan. The three numbers are one
46 result, reported at different seed counts and roundings.

47 **Supplementary Note 3: Parameters of the simulations**

48 Supplementary Table 1. Parameters of every simulation in this Article. The shared model layers are described in
49 the Methods of the companion Article.

Figure	Model	Parameters
1	melanoma-like	$k_{tx} = 3$, $k_{dm} = 0.1$, $k_{tl} = 0.4$, $k_{dp} = 0.02$ per h; net doubling 4 weeks, memory 5 net doublings, pre-resistant fraction 0.005 (1:200, above the traced 1:1,000 to 1:10,000 of [1]; Methods), $h_{max} = 0.0023$ per h, $EC_{50} = 0.3$, $m = 2$, protection $K = 150$, $q = 4$, growth arrest $IC_{50} = 0.3$, $m_g = 2$, with the same protection, $k_{off} \rightarrow k_{off}/(1 + 9d)$, fitness cost 0.5, partial protection: additional unprotected growth inhibition with $IC_{50} = 1$; 2,000 founders with promoter states drawn from the stationary distribution, 60 weeks, dt 4 h; progression from the nadir at 1.73 $\times$ the running minimum after the 8-week lead-in, loss of control at 1.2 $\times$ the pre-treatment size; cycle-gated variant (Supplementary Fig. 1) adds $\beta = 0.25$, $\varphi_0 = 0.5$, $w = 0.15$, and is repeated with $h_{max} = 0.0045$ per h so that the cycle-averaged hazard matches the cycle-blind one; the memory of the resistant state is scanned over 2, 3.5, 5, 8 and 12 net doublings with three seeds per point

Figure	Model	Parameters
2	MGMT model	same expression kinetics; net doubling 40 days, memory 4 net doublings, MGMT-expressing fraction 0.01 or 0.30, $h_{\max} = 0.03$ per h at peak, $EC_{50} = 0.4$ of the standard bolus peak, $m = 2$, $K = 150$, $q = 4$; elimination half-life 2.1 h; stoichiometric MGMT consumption at rate $k d V c / (c + K_m)$ with $k = 300$, $K_m = 150$, k chosen so that the consumption rate at the standard bolus peak matches the first-order parameterisation it replaces rather than fitted to data, and with no lesion pool as a state variable, so lesions are assumed to form in proportion to dose and to be repaired at once; 1,500 founders at the stationary promoter distribution, six 28-day cycles, dt 1 h

Supplementary Note 4: Clinical and laboratory reference data

Supplementary Table 2. Reference values used by the case studies (transcribed from the cited articles; files in `paper/data/`).

Quantity	Value	Source
Pre-resistant melanoma cells traced back from resistant fates	initial frequency ~1:1,000 to 1:10,000 (context, not a value used: the melanoma study here uses 1:200; Methods)	[1]
Pre-resistant melanoma cells	1:50 to 1:500 per marker; EGFR-high cells give 7.9 ± 0.9 fold more resistant colonies	[2]
N15-0385 glioblastoma doubling time	50 h	[3]
Temozolomide elimination half-life in plasma	1.8 h (single dose); 2.1 h (population model of plasma and cerebrospinal fluid; the value used here)	[4]; [5]

Quantity	Value	Source
Temozolomide penetration of the cerebrospinal fluid	exposure 20% of plasma exposure	[5]
MGMT / alkyltransferase depletion in peripheral blood mononuclear cells	−63% at 14 days, −73% at 21 days on protracted schedules; nadir 18.0 ± 2.26% of initial on a compressed 1,000 mg/m ² schedule	[6]; [7]
Tumour MGMT activity in orthotopic GBM43 xenografts	depleted by day 6 on both schedules; still suppressed at day 22 only on the 21-day schedule; back to baseline in both by day 29	[8]
RTOG 0525 regimens	150-200 mg/m ² days 1-5 vs 75-100 mg/m ² days 1-21 of 28-day cycles; median OS 16.6 vs 14.9 months	[9]
SWOG S1320 regimens	continuous vs 3 weeks off / 5 weeks on after an 8-week lead-in; median PFS 9.0 vs 5.5 months (HR 1.36 intermittent:continuous, $P = 0.063$; the trial pre-specified two-sided $\alpha = 0.2$ and 80% confidence intervals); median OS 29.2 months in both arms, a secondary end point the trial was not powered for	[10]

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